

Sensitivity Analysis of GE-LAV Predictions

Session 15 • When the framework is robust, when it isn't

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What we'll cover today

1

Why sensitivity matters

Calibration risk · model risk

2

Parameter sensitivity

κ , σ , $\bar{L}$ — local sensitivities

3

Crisis depth sensitivity

$L = -1.5$ vs -2.0 vs -2.5

4

Robustness checks

Five tests every analysis should pass

5

Practical sensitivity rules

When to trust, when to caveat

Why sensitivity matters for GE-LAV

CATEGORY 1

Highly robust predictions

Predictions where 10% parameter variation changes the result by $<5\%$. These can be cited with confidence.

→ Example: long-run mean valuation

CATEGORY 2

Moderately sensitive

Predictions where 10% parameter variation changes the result 5-15%. Cite with explicit caveats and ranges.

→ Example: optimal exit boundary

CATEGORY 3

Highly sensitive

Predictions where 10% parameter variation changes the result $>15\%$. Use ranges and qualitative direction, not point estimates.

→ Example: GFC-depth pricing

Knowing the category before using a result is essential for honest GE-LAV practice.

Parameter sensitivity table (calibrated baseline)

Each cell shows percentage change in the asset value for a $\pm 10\%$ change in that parameter.

Prediction	$\kappa \pm 10\%$	$\sigma \pm 10\%$	$\bar{L} \pm 10\%$	$L_0 \pm 10\%$
Long-run mean valuation	$\pm 1.5\%$	$\pm 2.0\%$	$\pm 3.0\%$	$\pm 0.0\%$
Term structure $\pi(L, T)$	$\pm 3.0\%$	$\pm 4.5\%$	$\pm 2.0\%$	$\pm 5.0\%$
Optimal exit boundary $L^*(t)$	$\pm 5.0\%$	$\pm 8.0\%$	$\pm 4.0\%$	$\pm 2.0\%$
Jensen bias magnitude	$\pm 4.0\%$	$\pm 10.0\%$	$\pm 1.0\%$	$\pm 6.0\%$
GFC-depth pricing ($L = -1.5$)	$\pm 8.0\%$	$\pm 18.0\%$	$\pm 5.0\%$	$\pm 28.0\%$
Welfare gap estimate	$\pm 6.0\%$	$\pm 15.0\%$	$\pm 4.0\%$	$\pm 22.0\%$
Pigouvian tax $\tau^*(L)$	$\pm 10.0\%$	$\pm 25.0\%$	$\pm 6.0\%$	$\pm 35.0\%$

Green: robust ($< 5\%$) • Gold: moderate (5-15%) • Red: highly sensitive ($> 15\%$)

Pattern: predictions about the tails (GFC, welfare, taxes) are most sensitive — exactly where the framework most claims to add value.

Five robustness tests for any GE-LAV analysis

Before presenting GE-LAV results, run these five tests:

1

Sub-sample stability

Does the calibration hold across 2003-2010 vs. 2011-2024?

If parameters differ by >25%, regime instability is a concern. Use sub-period results.

2

Out-of-sample prediction

Calibrate on 2003-2019. Predict 2020-2024. How close?

GE-LAV passes if predictions are within $\pm 20\%$ of observed values.

3

Alternate calibration sources

Use Lazard data → does it produce the same parameters as Evercore data?

Cross-source agreement is essential for credibility.

4

Bootstrap parameter uncertainty

Resample data 1,000 times. What's the 95% CI for each parameter?

Report results with bootstrap CIs, not just point estimates.

5

Stress-test alternative DGPs

What if $L(t)$ follows CIR instead of OU? Or jump-diffusion?

If results survive alternative DGP assumptions, robust. If not, model risk.

Session 15 summary

What we accomplished today

- 1 GE-LAV predictions vary widely in sensitivity — from <5% (robust) to >35% (highly sensitive)
- 2 Most robust: long-run mean valuation, term-structure shape
- 3 Most sensitive: tail predictions (GFC pricing, welfare, Pigouvian tax)
- 4 Run the five robustness tests before publishing or citing any GE-LAV result

Next session

Unit 4 begins: regulatory and platform implementation